

Case Assignment 5
BAN 630 – Optimization for Analytics

Group 3

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a. Based on historical data simulate the number of points Ryan would get in the remaining matchups (13-20). Compute the total points.

- Randomly samples from historical weekly points (with replacement) , Then Average Final Points (Simulated): 197 points

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Average Final Points: 197.56
Median Final Points: 198.00
Standard Deviation: 11.76
10th Percentile: 182.00
90th Percentile: 213.00
```

- There is 90% chance that Ryan will cross 182 points and 10% chance that Ryan will cross 213 end of season.
- Generates future points using a fitted normal distribution, Then Average Simulated Final Points: 197.6988934401993
 - Advantages: Captures underlying performance trends using mean and standard deviation and allows for natural randomness, including some unseen values.
 - Disadvantages: Assumes performance follows a normal distribution, which may not be the best fit.

b. Based on the last two years data (exhibit 3) and your simulation in a, find the probability that Ryan would make it to playoffs in 2018-19.

Simulated 8th Place Thresholds:

2016-17 Season - Mean: 191.26, 10th Percentile: 162.00, 90th Percentile: 207.00

2017-18 Season - Mean: 185.14, 10th Percentile: 170.00, 90th Percentile: 198.00

Probability of Ryan making playoffs based on 8th place mean:

2016-17 Season: 69.88%

2017-18 Season: 84.72%

Probability of Ryan making playoffs based on 10th percentile of 8th place:

2016-17 Season: 99.87%

2017-18 Season: 99.10%

Probability of Ryan making playoffs based on 90th percentile of 8th place:

2016-17 Season: 22.34%

2017-18 Season: 50.58%

- Based on 2016-17 Season final standings Ryan has 70% chance of making it to the playoff
- Based on 2017-18 Season final standings Ryan has 84.72% chance of making it to the playoff

c. Perform a sensitivity analysis of a and b by varying your assumptions and trying different scenarios.

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2016-17 Season: 22.34%

2017-18 Season: 50.58%

- With 162 points being the 8th position threshold (10th percentile of 2016-17 8th position simulation) there is 99.87% that Ryan qualifies for playoffs.
- With 170 points being the 8th position threshold (10th percentile of 2017-18 8th position simulation) there is 99.10% that Ryan qualifies for playoffs.
- With playoffs being very highly competitive With 198 points being the 8th position threshold (90th percentile of 2017-18 8th position simulation) there is 50% chance that Ryan makes it into the playoffs.

```
# Combine the historical data into one list for resampling
historical_weekly_points = ryan_2016_17 + ryan_2017_18*3
print(historical_weekly_points)
```

- As the recent season 2017-18 reflects slightly more on current season (changes or team combinations from current season) . Using 2016-17 , 2017-18 Ryan's performance in the ratio 1:3 . The average points that Ryan reaches is 195 points at end of the season.

d. Summarize your analysis in the form of key recommendations to Ryan.

- On Average Final Points (Simulated) that Ryan could reach is 197 points at the end of season.
- As there is trade deadline is until 17th match (matchup) Ryan can wait until match 16 points then decide on trading players or to focus on the playoffs. If the season is normal (threshold for 8th position being 190). If Ryan has 152 points by game 16 there is a 88% probability that me makes into the playoffs. Anything above 160 points by game 16 assures Ryan 97% of making into the playoffs.

- If the season is tight and there needs higher points for being in top 8 (now threshold being 195 points for 8th place finish)